(c) Change h in the range $10.0\,\text{cm} \leqslant h \leqslant 38.0\,\text{cm}$ and determine T. Repeat until you have six sets of values of h and T.

Record your results in a table. Include values of $\dfrac{S}{h}$ and T^2 in your table.

[8]

(d) **(i)** Plot a graph of T^2 on the y-axis against $\dfrac{S}{h}$ on the x-axis. [3]

(ii) Draw the straight line of best fit. [1]

(iii) Determine the gradient and y-intercept of this line.

gradient = ...

y-intercept = ...

[2]